Supplementary Data Tables

Decision Support System for Supplier Selection: A Comprehensive Approach

Shah Md Tasrifur Rahim, Syed Rizwanul Haque, Shahed Mahmud

Main Case Study Input Data (20 Criteria)

Table 1: Table 11: Averaged supplier performance scores used as diagonal GTMA values

Criteria Group	Sub-Criterion	Supplier A	Supplier B	Supplier C	Supplier D	Supplier E
Conventional	Cost	7.80	7.40	6.60	8.60	7.20
Conventional	Quality	7.60	9.40	7.60	5.40	8.40
Conventional	Delivery	8.80	7.00	7.80	5.60	6.80
Conventional	Lead Time	7.80	6.40	8.20	8.20	6.20
Conventional	Production Capacity	6.60	8.20	9.00	5.60	6.60
Environmental	Energy Consumption	5.80	8.00	7.20	4.20	7.20
Environmental	Waste Recycling	5.80	8.80	7.20	4.60	7.40
Environmental	Carbon Footprint	4.60	8.00	7.40	3.00	7.40
Environmental	Water Consumption	5.80	9.00	6.00	4.20	7.00
Environmental	Environmental Certification	6.60	9.20	8.60	4.60	6.00
Circular Economy	Product Recyclability	5.60	9.20	7.40	5.00	8.40
Circular Economy	Material Reuse	5.00	8.20	7.40	4.00	8.20
Circular Economy	Waste Recovery	5.60	9.00	6.80	3.80	7.60
Circular Economy	Recycled Material Usage	4.40	9.00	6.40	4.20	8.00
Circular Economy	Closed Loop Supply	5.20	8.40	6.80	3.80	7.40
Resilience	Risk Management	8.00	7.20	9.40	5.60	6.80
Resilience	Supply Redundancy	6.80	6.40	8.20	4.60	7.20
Resilience	Demand Surge Adaptability	8.00	7.00	9.20	5.20	6.20
Resilience	Inventory Buffer	6.80	6.60	8.40	5.20	7.00
Resilience	Supply Chain Visibility	8.00	7.20	9.00	4.60	6.80

Table 2: Table 12: Averaged criteria interdependency inputs (off-diagonal GTMA values)

Criteria Group	From Criterion	To Criterion	Input	Reverse
Conventional	Cost	Quality	5.80	4.20
Conventional	Cost	Delivery	5.00	5.00
Conventional	Cost	Lead Time	6.20	3.80
Conventional	Cost	Production Capacity	4.00	6.00
Conventional	Quality	Delivery	7.20	2.80
Conventional	Quality	Lead Time	5.40	4.60
Conventional	Quality	Production Capacity	6.00	4.00
Conventional	Delivery	Lead Time	7.60	2.40
Conventional	Delivery	Production Capacity	6.00	4.00
Conventional	Lead Time	Production Capacity	5.00	5.00
Environmental	Energy Consumption	Waste Recycling	6.60	3.40
Environmental	Energy Consumption	Carbon Footprint	8.20	1.80
Environmental	Energy Consumption	Water Consumption	6.20	3.80
Environmental	Energy Consumption	Environmental Certification	4.80	5.20

Table 2: Table 12: Averaged criteria interdependency inputs (off-diagonal GTMA values)

Criteria Group	From Criterion	To Criterion	Input	Reverse
Environmental	Waste Recycling	Carbon Footprint	5.20	4.80
Environmental	Waste Recycling	Water Consumption	5.00	5.00
Environmental	Waste Recycling	Environmental Certification	5.00	5.00
Environmental	Carbon Footprint	Water Consumption	4.00	6.00
Environmental	Carbon Footprint	Environmental Certification	6.80	3.20
Environmental	Water Consumption	Environmental Certification	5.00	5.00
Circular Economy	Product Recyclability	Material Reuse	7.60	2.40
Circular Economy	Product Recyclability	Waste Recovery	6.00	4.00
Circular Economy	Product Recyclability	Recycled Material Usage	8.00	2.00
Circular Economy	Product Recyclability	Closed Loop Supply	5.40	4.60
Circular Economy	Material Reuse	Waste Recovery	6.60	3.40
Circular Economy	Material Reuse	Recycled Material Usage	8.20	1.80
Circular Economy	Material Reuse	Closed Loop Supply	5.40	4.60
Circular Economy	Waste Recovery	Recycled Material Usage	7.00	3.00
Circular Economy	Waste Recovery	Closed Loop Supply	6.00	4.00
Circular Economy	Recycled Material Usage	Closed Loop Supply	6.80	3.20
Resilience	Risk Management	Supply Redundancy	7.20	2.80
Resilience	Risk Management	Demand Surge Adaptability	6.00	4.00
Resilience	Risk Management	Inventory Buffer	5.80	4.20
Resilience	Risk Management	Supply Chain Visibility	6.60	3.40
Resilience	Supply Redundancy	Demand Surge Adaptability	6.20	3.80
Resilience	Supply Redundancy	Inventory Buffer	8.60	1.40
Resilience	Supply Redundancy	Supply Chain Visibility	7.20	2.80
Resilience	Demand Surge Adaptability	Inventory Buffer	6.20	3.80
Resilience	Demand Surge Adaptability	Supply Chain Visibility	6.20	3.80
Resilience	Inventory Buffer	Supply Chain Visibility	5.60	4.40

Table 3: Table 13: Category-level interdependency values used in the final synthesis matrix

From Criteria Group	To Criteria Group	Input	Reverse
Conventional	Environmental	4.80	5.20
Conventional	Circular Economy	4.40	5.60
Conventional	Resilience	5.80	4.20
Environmental	Circular Economy	8.20	1.80
Environmental	Resilience	5.20	4.80
Circular Economy	Resilience	5.80	4.20

Main Case Study Sensitivity Summary

Table 4: Table 16: Sensitivity analysis results for the main case study

Sensitivity Type	Change	Top Supplier	Rank Change	Spearman Correlation
Diagonal scores	-20%	Supplier B	No change	1.000
Diagonal scores	-10%	Supplier B	No change	1.000
Diagonal scores	$+10\%$	Supplier B	No change	1.000
Diagonal scores	$+20\%$	Supplier B	No change	1.000
Off-diagonal values	-20%	Supplier B	No change	1.000
Off-diagonal values	-10%	Supplier B	No change	1.000
Off-diagonal values	$+10\%$	Supplier B	No change	1.000
Off-diagonal values	$+20\%$	Supplier B	No change	1.000
Simultaneous adjustment	-20%	Supplier B	No change	1.000
Simultaneous adjustment	-10%	Supplier B	No change	1.000
Simultaneous adjustment	$+10\%$	Supplier B	No change	1.000
Simultaneous adjustment	$+20\%$	Supplier B	No change	1.000

Full-Scale Application Input Data (43 Criteria)

Table 5: Table 17: Supplier performance scores used as diagonal GTMA values (full-scale, 43 criteria)

Criteria Group	Sub-Criterion	Supplier A	Supplier B	Supplier C	Supplier D	Supplier E
Conventional	Cost	7.80	7.40	6.60	8.60	7.20
Conventional	Quality	7.60	9.40	7.60	5.40	8.40
Conventional	Delivery	8.80	7.00	7.80	5.60	6.80
Conventional	Lead Time	7.80	6.40	8.20	8.20	6.20
Conventional	Production Capacity	6.60	8.20	9.00	5.60	6.60
Conventional	Financial Stability	7.40	7.80	8.60	5.60	7.00
Conventional	Technical Capability	7.20	8.20	9.00	5.40	7.60
Conventional	After Sales Service	7.00	8.40	8.20	5.20	7.80
Conventional	Market Reputation	7.60	8.60	8.40	5.00	7.40
Conventional	Supplier Experience	7.40	8.00	8.80	5.80	7.20
Conventional	Warranty Policy	6.80	8.00	7.80	5.20	7.60
Conventional	Inventory Capability	6.80	7.00	8.40	5.60	8.20
Conventional	Communication Efficiency	7.40	7.60	8.60	5.20	8.00
Conventional	Flexibility	7.00	7.00	8.60	5.00	8.60
Environmental	Energy Consumption	5.80	8.00	7.20	4.20	7.20
Environmental	Waste Recycling	5.80	8.80	7.20	4.60	7.40
Environmental	Carbon Footprint	4.60	8.00	7.40	3.00	7.40
Environmental	Water Consumption	5.80	9.00	6.00	4.20	7.00
Environmental	Green Packaging	6.20	8.80	7.20	4.00	7.60
Environmental	Pollution Emission	5.00	8.20	7.20	3.40	7.20
Environmental	Renewable Energy Usage	5.60	9.00	7.00	3.60	6.80
Environmental	Hazardous Material Handling	6.40	8.40	7.80	4.20	6.60
Environmental	Environmental Certification	6.60	9.20	8.60	4.60	6.00
Environmental	Eco Friendly Design	6.00	9.00	7.40	4.00	7.80
Circular Economy	Product Recyclability	5.60	9.20	7.40	5.00	8.40
Circular Economy	Reverse Logistics	5.40	8.60	7.00	4.00	8.80
Circular Economy	Material Reuse	5.00	8.20	7.40	4.00	8.20
Circular Economy	Remanufacturing Capability	5.00	8.40	6.80	3.60	8.20
Circular Economy	Waste Recovery	5.60	9.00	6.80	3.80	7.60
Circular Economy	Closed Loop Supply	5.20	8.40	6.80	3.80	7.40
Circular Economy	Recycled Material Usage	4.40	9.00	6.40	4.20	8.00
Circular Economy	Product Life Extension	5.60	8.80	7.20	4.20	8.60
Circular Economy	Modular Design	5.20	8.20	7.60	4.00	8.40
Resilience	Risk Management	8.00	7.20	9.40	5.60	6.80
Resilience	Supply Redundancy	6.80	6.40	8.20	4.60	7.20
Resilience	Disaster Recovery Plan	7.80	7.00	9.20	5.00	6.80
Resilience	Demand Surge Adaptability	8.00	7.00	9.20	5.20	6.20
Resilience	Supplier Diversification	7.00	6.80	8.60	5.20	7.40
Resilience	Inventory Buffer	6.80	6.60	8.40	5.20	7.00
Resilience	Cyber Security Readiness	7.40	7.00	8.80	4.80	6.60
Resilience	Digital Tracking Capability	7.60	7.40	9.00	4.60	7.00
Resilience	Emergency Response	8.00	7.00	9.20	5.20	7.00
Resilience	Supply Chain Visibility	8.00	7.20	9.00	4.60	6.80

Table 6: Table 22: Category-level interdependency values used in the 4×4 synthesis matrix (full-scale)

From Criteria Group	To Criteria Group	Input	Reverse
Conventional	Environmental	4.80	5.20
Conventional	Circular Economy	4.40	5.60
Conventional	Resilience	5.80	4.20
Environmental	Circular Economy	8.20	1.80
Environmental	Resilience	5.20	4.80
Circular Economy	Resilience	5.80	4.20

Table 7: Table 18: Criteria interdependency values — Conventional Criteria (14 sub-criteria, 91 upper-triangle pairs)

From Criterion	To Criterion	Input (r_{ij})	Reverse (r_{ji})
Cost	Quality	5.80	4.20
Cost	Delivery	5.00	5.00
Cost	Lead Time	6.20	3.80
Cost	Production Capacity	4.00	6.00
Cost	Financial Stability	5.80	4.20
Cost	Technical Capability	4.60	5.40
Cost	After Sales Service	6.40	3.60
Cost	Market Reputation	4.40	5.60
Cost	Supplier Experience	5.80	4.20
Cost	Warranty Policy	6.40	3.60
Cost	Inventory Capability	4.80	5.20
Cost	Communication Efficiency	5.80	4.20
Cost	Flexibility	5.80	4.20
Quality	Delivery	7.20	2.80
Quality	Lead Time	5.40	4.60
Quality	Production Capacity	6.00	4.00
Quality	Financial Stability	4.60	5.40
Quality	Technical Capability	7.40	2.60
Quality	After Sales Service	5.40	4.60
Quality	Market Reputation	5.60	4.40
Quality	Supplier Experience	4.80	5.20
Quality	Warranty Policy	5.40	4.60
Quality	Inventory Capability	6.00	4.00
Quality	Communication Efficiency	4.80	5.20
Quality	Flexibility	4.80	5.20
Delivery	Lead Time	7.60	2.40
Delivery	Production Capacity	6.00	4.00
Delivery	Financial Stability	5.20	4.80
Delivery	Technical Capability	6.40	3.60
Delivery	After Sales Service	6.00	4.00
Delivery	Market Reputation	6.20	3.80
Delivery	Supplier Experience	5.40	4.60
Delivery	Warranty Policy	6.00	4.00
Delivery	Inventory Capability	6.80	3.20
Delivery	Communication Efficiency	5.40	4.60
Delivery	Flexibility	5.40	4.60
Lead Time	Production Capacity	5.00	5.00
Lead Time	Financial Stability	5.60	4.40
Lead Time	Technical Capability	4.60	5.40
Lead Time	After Sales Service	6.40	3.60
Lead Time	Market Reputation	4.40	5.60
Lead Time	Supplier Experience	5.80	4.20
Lead Time	Warranty Policy	6.40	3.60
Lead Time	Inventory Capability	4.80	5.20
Lead Time	Communication Efficiency	5.80	4.20
Lead Time	Flexibility	6.40	3.60
Production Capacity	Financial Stability	6.80	3.20
Production Capacity	Technical Capability	7.00	3.00
Production Capacity	After Sales Service	5.00	5.00
Production Capacity	Market Reputation	5.20	4.80
Production Capacity	Supplier Experience	4.40	5.60
Production Capacity	Warranty Policy	5.00	5.00
Production Capacity	Inventory Capability	5.60	4.40
Production Capacity	Communication Efficiency	4.40	5.60
Production Capacity	Flexibility	4.40	5.60
Financial Stability	Technical Capability	4.40	5.60
Financial Stability	After Sales Service	6.20	3.80
Financial Stability	Market Reputation	6.40	3.60
Financial Stability	Supplier Experience	5.60	4.40
Financial Stability	Warranty Policy	6.20	3.80
Financial Stability	Inventory Capability	4.60	5.40

From Criterion	To Criterion	Input (r_{ij})	Reverse (r_{ji})
Financial Stability	Communication Efficiency	5.60	4.40
Financial Stability	Flexibility	5.60	4.40
Technical Capability	After Sales Service	5.20	4.80
Technical Capability	Market Reputation	5.40	4.60
Technical Capability	Supplier Experience	4.60	5.40
Technical Capability	Warranty Policy	5.20	4.80
Technical Capability	Inventory Capability	5.80	4.20
Technical Capability	Communication Efficiency	4.60	5.40
Technical Capability	Flexibility	4.60	5.40
After Sales Service	Market Reputation	5.00	5.00
After Sales Service	Supplier Experience	6.40	3.60
After Sales Service	Warranty Policy	7.20	2.80
After Sales Service	Inventory Capability	5.40	4.60
After Sales Service	Communication Efficiency	6.40	3.60
After Sales Service	Flexibility	6.40	3.60
Market Reputation	Supplier Experience	7.00	3.00
Market Reputation	Warranty Policy	5.00	5.00
Market Reputation	Inventory Capability	5.60	4.40
Market Reputation	Communication Efficiency	4.40	5.60
Market Reputation	Flexibility	4.40	5.60
Supplier Experience	Warranty Policy	6.40	3.60
Supplier Experience	Inventory Capability	4.80	5.20
Supplier Experience	Communication Efficiency	5.80	4.20
Supplier Experience	Flexibility	5.80	4.20
Warranty Policy	Inventory Capability	5.40	4.60
Warranty Policy	Communication Efficiency	6.40	3.60
Warranty Policy	Flexibility	6.40	3.60
Inventory Capability	Communication Efficiency	4.80	5.20
Inventory Capability	Flexibility	4.80	5.20
Communication Efficiency	Flexibility	7.40	2.60

Table 8: Table 19: Criteria interdependency values — Environmental Criteria (10 sub-criteria, 45 upper-triangle pairs)

From Criterion	To Criterion	Input (r_{ij})	Reverse (r_{ji})
Energy Consumption	Waste Recycling	6.60	3.40
Energy Consumption	Carbon Footprint	8.20	1.80
Energy Consumption	Water Consumption	6.20	3.80
Energy Consumption	Green Packaging	5.20	4.80
Energy Consumption	Pollution Emission	5.20	4.80
Energy Consumption	Renewable Energy Usage	7.40	2.60
Energy Consumption	Hazardous Material Handling	5.20	4.80
Energy Consumption	Environmental Certification	4.80	5.20
Energy Consumption	Eco Friendly Design	5.80	4.20
Waste Recycling	Carbon Footprint	5.20	4.80
Waste Recycling	Water Consumption	5.00	5.00
Waste Recycling	Green Packaging	6.20	3.80
Waste Recycling	Pollution Emission	6.40	3.60
Waste Recycling	Renewable Energy Usage	6.20	3.80
Waste Recycling	Hazardous Material Handling	6.40	3.60
Waste Recycling	Environmental Certification	5.00	5.00
Waste Recycling	Eco Friendly Design	4.80	5.20
Carbon Footprint	Water Consumption	4.00	6.00
Carbon Footprint	Green Packaging	5.80	4.20
Carbon Footprint	Pollution Emission	8.00	2.00
Carbon Footprint	Renewable Energy Usage	5.60	4.40
Carbon Footprint	Hazardous Material Handling	5.80	4.20
Carbon Footprint	Environmental Certification	6.80	3.20
Carbon Footprint	Eco Friendly Design	6.40	3.60
Water Consumption	Green Packaging	6.60	3.40
Water Consumption	Pollution Emission	5.80	4.20

From Criterion	To Criterion	Input (r_{ij})	Reverse (r_{ji})
Water Consumption	Renewable Energy Usage	6.40	3.60
Water Consumption	Hazardous Material Handling	6.60	3.40
Water Consumption	Environmental Certification	5.00	5.00
Water Consumption	Eco Friendly Design	5.00	5.00
Green Packaging	Pollution Emission	6.00	4.00
Green Packaging	Renewable Energy Usage	5.80	4.20
Green Packaging	Hazardous Material Handling	6.00	4.00
Green Packaging	Environmental Certification	5.40	4.60
Green Packaging	Eco Friendly Design	7.20	2.80
Pollution Emission	Renewable Energy Usage	5.80	4.20
Pollution Emission	Hazardous Material Handling	6.00	4.00
Pollution Emission	Environmental Certification	5.40	4.60
Pollution Emission	Eco Friendly Design	6.60	3.40
Renewable Energy Usage	Hazardous Material Handling	5.80	4.20
Renewable Energy Usage	Environmental Certification	5.20	4.80
Renewable Energy Usage	Eco Friendly Design	6.40	3.60
Hazardous Material Handling	Environmental Certification	7.60	2.40
Hazardous Material Handling	Eco Friendly Design	6.60	3.40
Environmental Certification	Eco Friendly Design	6.80	3.20

Table 9: Table 20: Criteria interdependency values — Circular Economy Criteria (9 sub-criteria, 36 upper-triangle pairs)

From Criterion	To Criterion	Input (r_{ij})	Reverse (r_{ji})
Product Recyclability	Reverse Logistics	5.20	4.80
Product Recyclability	Material Reuse	7.60	2.40
Product Recyclability	Remanufacturing Capability	6.00	4.00
Product Recyclability	Waste Recovery	6.00	4.00
Product Recyclability	Closed Loop Supply	5.40	4.60
Product Recyclability	Recycled Material Usage	8.00	2.00
Product Recyclability	Product Life Extension	6.00	4.00
Product Recyclability	Modular Design	5.40	4.60
Reverse Logistics	Material Reuse	6.00	4.00
Reverse Logistics	Remanufacturing Capability	5.60	4.40
Reverse Logistics	Waste Recovery	5.80	4.20
Reverse Logistics	Closed Loop Supply	8.20	1.80
Reverse Logistics	Recycled Material Usage	6.60	3.40
Reverse Logistics	Product Life Extension	6.40	3.60
Reverse Logistics	Modular Design	5.00	5.00
Material Reuse	Remanufacturing Capability	7.60	2.40
Material Reuse	Waste Recovery	6.60	3.40
Material Reuse	Closed Loop Supply	5.40	4.60
Material Reuse	Recycled Material Usage	8.20	1.80
Material Reuse	Product Life Extension	6.80	3.20
Material Reuse	Modular Design	6.20	3.80
Remanufacturing Capability	Waste Recovery	6.60	3.40
Remanufacturing Capability	Closed Loop Supply	5.20	4.80
Remanufacturing Capability	Recycled Material Usage	5.20	4.80
Remanufacturing Capability	Product Life Extension	6.40	3.60
Remanufacturing Capability	Modular Design	5.80	4.20
Waste Recovery	Closed Loop Supply	6.00	4.00
Waste Recovery	Recycled Material Usage	7.00	3.00
Waste Recovery	Product Life Extension	6.60	3.40
Waste Recovery	Modular Design	6.00	4.00
Closed Loop Supply	Recycled Material Usage	3.20	6.80
Closed Loop Supply	Product Life Extension	5.20	4.80
Closed Loop Supply	Modular Design	6.80	3.20
Recycled Material Usage	Product Life Extension	5.20	4.80
Recycled Material Usage	Modular Design	6.80	3.20
Product Life Extension	Modular Design	7.80	2.20

Table 10: Table 21: Criteria interdependency values — Resilience Criteria (10 sub-criteria, 45 upper-triangle pairs)

From Criterion	**To Criterion**	**Input (r_{ij})**	**Reverse (r_{ji})**
Risk Management	Supply Redundancy	7.20	2.80
Risk Management	Disaster Recovery Plan	7.80	2.20
Risk Management	Demand Surge Adaptability	6.00	4.00
Risk Management	Supplier Diversification	6.50	3.50
Risk Management	Inventory Buffer	5.80	4.20
Risk Management	Cyber Security Readiness	4.70	5.30
Risk Management	Digital Tracking Capability	4.70	5.30
Risk Management	Emergency Response	6.80	3.20
Risk Management	Supply Chain Visibility	6.60	3.40
Supply Redundancy	Disaster Recovery Plan	5.90	4.10
Supply Redundancy	Demand Surge Adaptability	6.20	3.80
Supply Redundancy	Supplier Diversification	7.60	2.40
Supply Redundancy	Inventory Buffer	8.60	1.40
Supply Redundancy	Cyber Security Readiness	6.10	3.90
Supply Redundancy	Digital Tracking Capability	6.10	3.90
Supply Redundancy	Emergency Response	5.70	4.30
Supply Redundancy	Supply Chain Visibility	7.20	2.80
Disaster Recovery Plan	Demand Surge Adaptability	6.50	3.50
Disaster Recovery Plan	Supplier Diversification	5.10	4.90
Disaster Recovery Plan	Inventory Buffer	5.10	4.90
Disaster Recovery Plan	Cyber Security Readiness	5.50	4.50
Disaster Recovery Plan	Digital Tracking Capability	5.50	4.50
Disaster Recovery Plan	Emergency Response	8.00	2.00
Disaster Recovery Plan	Supply Chain Visibility	6.50	3.50
Demand Surge Adaptability	Supplier Diversification	6.30	3.70
Demand Surge Adaptability	Inventory Buffer	6.20	3.80
Demand Surge Adaptability	Cyber Security Readiness	6.70	3.30
Demand Surge Adaptability	Digital Tracking Capability	6.70	3.30
Demand Surge Adaptability	Emergency Response	6.30	3.70
Demand Surge Adaptability	Supply Chain Visibility	6.20	3.80
Supplier Diversification	Inventory Buffer	4.90	5.10
Supplier Diversification	Cyber Security Readiness	5.30	4.70
Supplier Diversification	Digital Tracking Capability	5.30	4.70
Supplier Diversification	Emergency Response	4.90	5.10
Supplier Diversification	Supply Chain Visibility	6.30	3.70
Inventory Buffer	Cyber Security Readiness	5.30	4.70
Inventory Buffer	Digital Tracking Capability	5.30	4.70
Inventory Buffer	Emergency Response	4.90	5.10
Inventory Buffer	Supply Chain Visibility	5.60	4.40
Cyber Security Readiness	Digital Tracking Capability	7.40	2.60
Cyber Security Readiness	Emergency Response	5.30	4.70
Cyber Security Readiness	Supply Chain Visibility	6.70	3.30
Digital Tracking Capability	Emergency Response	5.30	4.70
Digital Tracking Capability	Supply Chain Visibility	7.80	2.20
Emergency Response	Supply Chain Visibility	6.30	3.70

Full-Scale Application Sensitivity Results

Table 11: Table 25: Full-scale sensitivity analysis results based on final SPI values

Scenario	Supplier B	Supplier C	Supplier E	Supplier A	Supplier D
Base case	3.145×10^{60}	2.825×10^{60}	1.939×10^{60}	9.102×10^{59}	2.179×10^{59}
Diagonal −20%	8.603×10^{59}	7.896×10^{59}	5.843×10^{59}	3.191×10^{59}	1.017×10^{59}
Diagonal −10%	1.645×10^{60}	1.494×10^{60}	1.064×10^{60}	5.389×10^{59}	1.488×10^{59}
Diagonal +10%	5.954×10^{60}	5.270×10^{60}	3.532×10^{60}	1.537×10^{60}	3.189×10^{59}
Diagonal +20%	9.413×10^{60}	8.329×10^{60}	6.250×10^{60}	2.575×10^{60}	4.647×10^{59}
Off-diagonal −20%	4.783×10^{60}	4.307×10^{60}	2.958×10^{60}	1.408×10^{60}	3.428×10^{59}
Off-diagonal −10%	4.669×10^{60}	4.201×10^{60}	2.889×10^{60}	1.371×10^{60}	3.333×10^{59}
Off-diagonal +10%	1.416×10^{60}	1.269×10^{60}	8.658×10^{59}	3.984×10^{59}	9.237×10^{58}
Off-diagonal +20%	4.000×10^{59}	3.568×10^{59}	2.410×10^{59}	1.074×10^{59}	2.362×10^{58}
Simultaneous −20%	1.328×10^{60}	1.221×10^{60}	9.041×10^{59}	4.991×10^{59}	1.613×10^{59}
Simultaneous −10%	2.459×10^{60}	2.236×10^{60}	1.596×10^{60}	8.162×10^{59}	2.286×10^{59}
Simultaneous +10%	2.719×10^{60}	2.399×10^{60}	1.598×10^{60}	6.803×10^{59}	1.363×10^{59}
Simultaneous +20%	1.275×10^{60}	1.119×10^{60}	8.313×10^{59}	3.222×10^{59}	5.257×10^{58}
Diag −10%, off-diag +10%	7.305×10^{59}	6.616×10^{59}	4.691×10^{59}	2.333×10^{59}	6.260×10^{58}
Diag +10%, off-diag −10%	8.779×10^{60}	7.785×10^{60}	5.228×10^{60}	2.303×10^{60}	4.861×10^{59}

In all 14 scenarios above, Top Supplier = Supplier B with No Rank Change, and Spearman correlation = 1.000. See Table 26 for the summary.

Table 12: Table 26: Rank-stability summary for the full-scale sensitivity analysis

Sensitivity Group	Number of Scenarios	Rank Reversal	Spearman Correlation
Diagonal adjustment	4	0	1.000
Off-diagonal adjustment	4	0	1.000
Same-direction simultaneous adjustment	4	0	1.000
Opposite-direction simultaneous adjustment	2	0	1.000
Total	14	0	1.000